ReligionBERT: Domain-Adaptive Pre-Training of BERT on Biblical Corpora for Religious NLP Tasks

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Abstract. Pre-trained language models such as BERT achieve strong performance on general NLP benchmarks but struggle with the distinctive vocabulary, archaic syntax, and dense theological symbolism found in religious texts. We present ReligionBERT and MultiReligionBERT, two domain-adapted BERT models produced by continued masked language modelling pre-training on the Bible corpus across monolingual English and 12-language multilingual settings. Alongside these models, we introduce three automatically constructed datasets: a 21,994-pair semantic similarity dataset, a 7,726-sample 66-class book classification dataset, and a 1,199-example LLM-assisted extractive QA dataset verified at 100% quality on a 200-sample human review. Benchmarking across four model variants and three task types shows that ReligionBERT outperforms generic BERT on five of six metrics, with the strongest gain of +7.50 Exact Match points on extractive QA. Perplexity on held-out Bible text falls from 15.48 for generic BERT to 3.73 for ReligionBERT, a 75.9% reduction. Zero-shot cross-lingual transfer on five African languages demonstrates that multilingual domain-specific pre-training confers measurable advantages over mBERT, including a result on Ewe where mBERT scores 0.00% accuracy and MultiReligionBERT scores 3.37%. Both models and all datasets are released publicly on HuggingFace.

Keywords: domain-adaptive pre-training, BERT, biblical corpus, religious NLP, cross-lingual transfer, African languages, extractive QA

1 Introduction

Pre-trained language models (PLMs) trained on large general-purpose corpora have reshaped natural language processing across a wide range of tasks [1, 2]. Their representations reflect the distributional properties of those corpora, which consist predominantly of contemporary, secular text. Religious texts occupy a distinctly different linguistic register: they employ archaic grammatical constructions, dense metaphor and allegory, domain-specific theological terminology, and long-range intertextual dependencies that general corpora do not capture adequately [5].

Domain-adaptive pre-training (DAPT), in which a PLM undergoes continued pre-training on in-domain text before task-specific fine-tuning, has proven effective in biomedical [4], legal [2], and scientific text [1, 6] settings. Religious NLP represents an underexplored application of this paradigm. The Bible corpus provides a freely available, structurally consistent, verse-aligned parallel corpus spanning more than 100 languages, making it well-suited to multi-task and cross-lingual studies.

This paper makes the following contributions:

- ReligionBERT and MultiReligionBERT are produced via continued MLM pre-training on monolingual English and 12-language multilingual Bible corpora and released publicly on HuggingFace.
- Three automatically constructed datasets for religious NLP are introduced: a 21,994-pair semantic similarity dataset, a 7,726-sample 66-class book classification dataset, and a

1,199-example LLM-assisted extractive QA dataset with 100% verified annotation quality.
- The first systematic multi-task, multi-model benchmarking study of Bible corpus domain adaptation is provided across four model variants and three task types.
- The first empirical study of zero-shot cross-lingual transfer for African language religious NLP using domain-adapted multilingual models is presented, covering five languages including the severely low-resource Ewe.

2 Related Work

2.1 Domain-Adaptive Pre-Training

The DAPT paradigm formalised by Gururangan et al. [6] demonstrated that continued pre-training on in-domain text consistently improves downstream performance. BioBERT [4] and SciBERT [1] applied this approach to biomedical and scientific literature respectively. LegalBERT [2] extended the pattern to the legal domain. To the best of our knowledge, this is the first systematic application of the DAPT paradigm to religious corpora involving multi-task benchmarking and cross-lingual transfer to low-resource African languages

2.2 Multilingual Models and Cross-Lingual Transfer

Multilingual BERT [3] demonstrated zero-shot cross-lingual transfer across 104 languages without target-language fine-tuning. XLM-RoBERTa [7] improved substantially on mBERT by training on 2.5 terabytes of multilingual CommonCrawl text. Neither incorporates domain-specific pre-training. MultiReligionBERT combines multilingual pre-training with religious domain adaptation, targeting cross-lingual religious NLP for low-resource African languages.

2.3 Religious and Biblical Text Corpora

The Bible corpus exists in hundreds of translations across diverse languages, making it one of the most multilingual parallel corpora available [10]. Early computational work applied keyword search and topic modelling to biblical corpora [11]. More recent work has explored general-purpose LLMs for religious question answering [14]. No prior work has performed continued BERT pre-training on biblical corpora with systematic multi-task benchmarking and cross-lingual transfer evaluation for African languages.

Table 1. Comparative summary of related studies in domain adaptation for specialised text.

Authors	Domain	Method	Key Finding	Gap Addressed Here
Gururangan et al. [6]	General	DAPT on domain corpora	Consistent gains across 4 domains	Religious domain; cross-lingual
Lee et al. [4]	Biomedical	BioBERT pre-training	Strong biomedical NLP gains	Religious setting; African languages
Beltagy et al. [1]	Scientific	SciBERT from scratch	State-of-the-art scientific tasks	Multilingual; low-resource
Chalkidis et al. [2]	Legal	LegalBERT pre-training	Improved legal NLP tasks	Religious domain; multilingual
Conneau et al. [7]	Multilingual	XLM-RoBERTa large-scale	Strong cross-lingual transfer	Domain-specific religious adaptation
This study	Religious (Bible)	DAPT + multi-task	Gains on 5/6 metrics; African transfer	All of the above

3 Data

3.1 Pre-Training Corpus

The primary corpus was sourced from the christos-c/bible-corpus repository, which contains 108 XML-formatted Bible translations with verse-level structure. For monolingual pre-training (ReligionBERT), the King James Version and World English Bible were concatenated, yielding 62,197 verses. For multilingual pre-training (MultiReligionBERT), twelve languages were selected: English, French, Spanish, Portuguese, German, Amharic, Shona, Xhosa, Malagasy, Somali, and Zarma, producing 372,652 verses. Ewe and Swahili New Testament-only translations were retained for cross-lingual evaluation only. A RAM-exhaustion issue during multilingual tokenisation was resolved by processing the corpus in 50,000-line chunks saved as HuggingFace dataset shards. Table 2 summarises corpus statistics.

Table 2. Pre-training corpus statistics.

Corpus	Languages	Verses	Base Model
English Bible (KJV + WEB)	1 (English)	62,197	bert-base-uncased
Multilingual Bible	12	372,652	bert-base-multilingual-cased

3.2 Fine-Tuning Dataset Construction

Three task-specific datasets were constructed automatically from the Bible corpus. Figure 1 illustrates the full pipeline. All datasets used an 80/10/10 train/validation/test split.

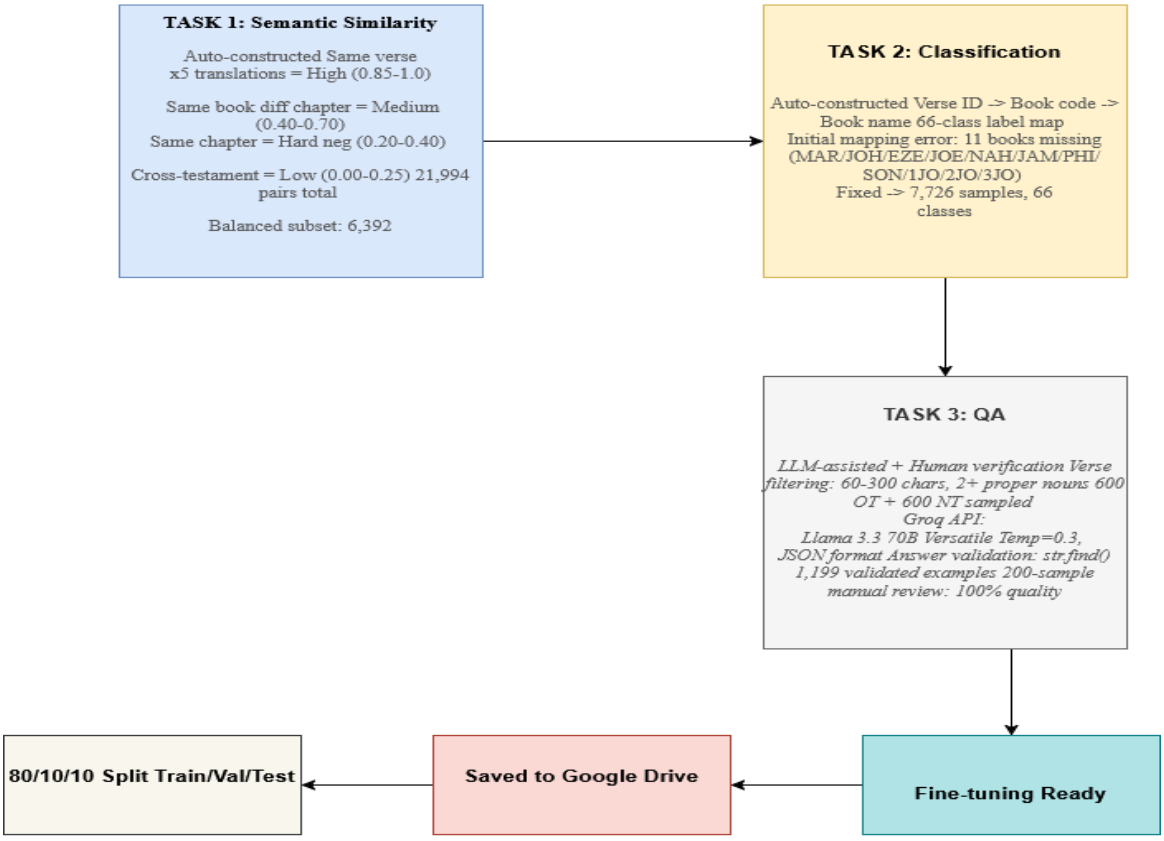

Fig. 1. Automatic and LLM-assisted dataset construction pipeline for the three fine-tuning tasks.

3.2.1 Semantic Similarity. Verse pairs were assigned similarity scores using the structural properties of the parallel corpus. Same-verse cross-translation pairs across five language pairs

received scores of 0.85 to 1.00, yielding 15,000 high-similarity pairs. Same-book different-chapter pairs received 0.40 to 0.70; same-chapter pairs received 0.20 to 0.40 (hard negatives); cross-testament pairs received 0.00 to 0.25. The full dataset contained 21,994 pairs. A balanced subset of 6,392 pairs was produced by downsampling the dominant class for staged training.

3.2.2 Book Classification. Each verse was labelled with its source book, producing a 66-class dataset. An initial run produced only 55 classes due to eleven non-standard book codes in the XML corpus (e.g., MAR for MRK, JOH for JHN, PHI for PHP, 1JO/2JO/3JO for 1JN/2JN/3JN). All mappings were corrected and the dataset regenerated, yielding 7,726 samples across all 66 books.

3.2.3 Extractive Question Answering. A dataset of 1,199 examples was generated using Llama 3.3 70B Versatile via the Groq API (temperature 0.3, JSON response format). Verses between 60 and 300 characters containing at least two proper nouns were selected, balanced across approximately 600 Old Testament and 600 New Testament passages. Each verse was submitted with a structured prompt requiring a factual question and an answer that is a verbatim substring of the verse. Answer validity was verified by exact substring matching. A 200-sample human review found zero quality issues (100% verified quality). The dataset is formatted in SQuAD v2-compatible JSON [8].

Table 3. Semantic similarity dataset statistics.

Category	Pairs	Score Range	Construction Method
High similarity	15,000	0.85-1.00	Same verse, five cross-translation pairs
Low similarity	4,000	0.00-0.25	Cross-testament OT-NT random pairs
Medium similarity	1,554	0.40-0.70	Same book, different chapters
Hard negative	1,440	0.20-0.40	Same chapter, adjacent verses
Full dataset	21,994	0.00-1.00	All categories combined
Balanced subset	6,392	0.00-1.00	Downsampled for class balance

Table 4. Extractive QA dataset properties.

Property	Value
Total examples	1,199
Generation model	Llama 3.3 70B Versatile (Groq API, temperature 0.3)
Validation method	Exact substring matching (Python str.find)
Human review sample	200 examples
Issues found	0 (100% verified quality)
Train / Validation / Test	959 / 120 / 120
Format	SQuAD v2 compatible JSON

4 Models and Pre-Training

4.1 Base Models

Two base models were used. bert-base-uncased is a 109M-parameter English BERT model with a 30,522-token WordPiece vocabulary pre-trained on English Wikipedia and BooksCorpus. bert-base-multilingual-cased (mBERT) is a 178M-parameter model pre-trained on Wikipedia text from 104 languages. These bases produced ReligionBERT and MultiReligionBERT respectively and served as the primary baselines. XLM-RoBERTa [7] was used as an additional cross-lingual baseline. Figure 2 shows the BERT architecture; Figure 3 shows the full experimental pipeline across all five research objectives.

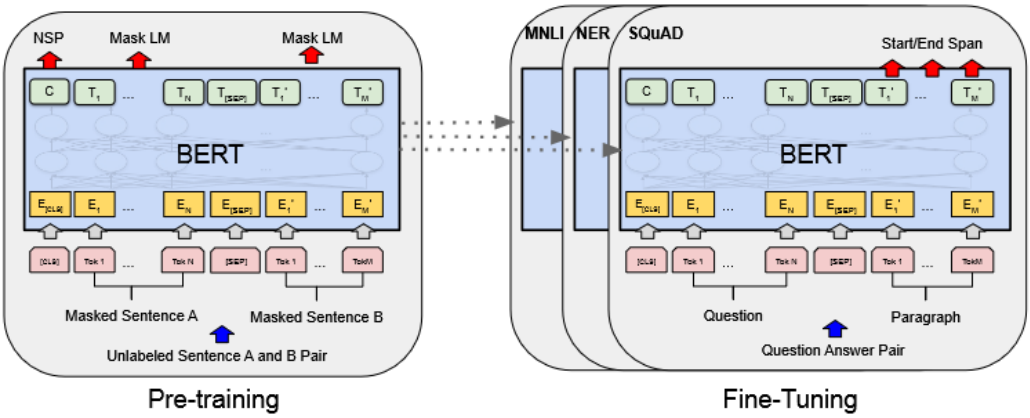

Fig. 2. BERT pre-training and fine-tuning architecture.

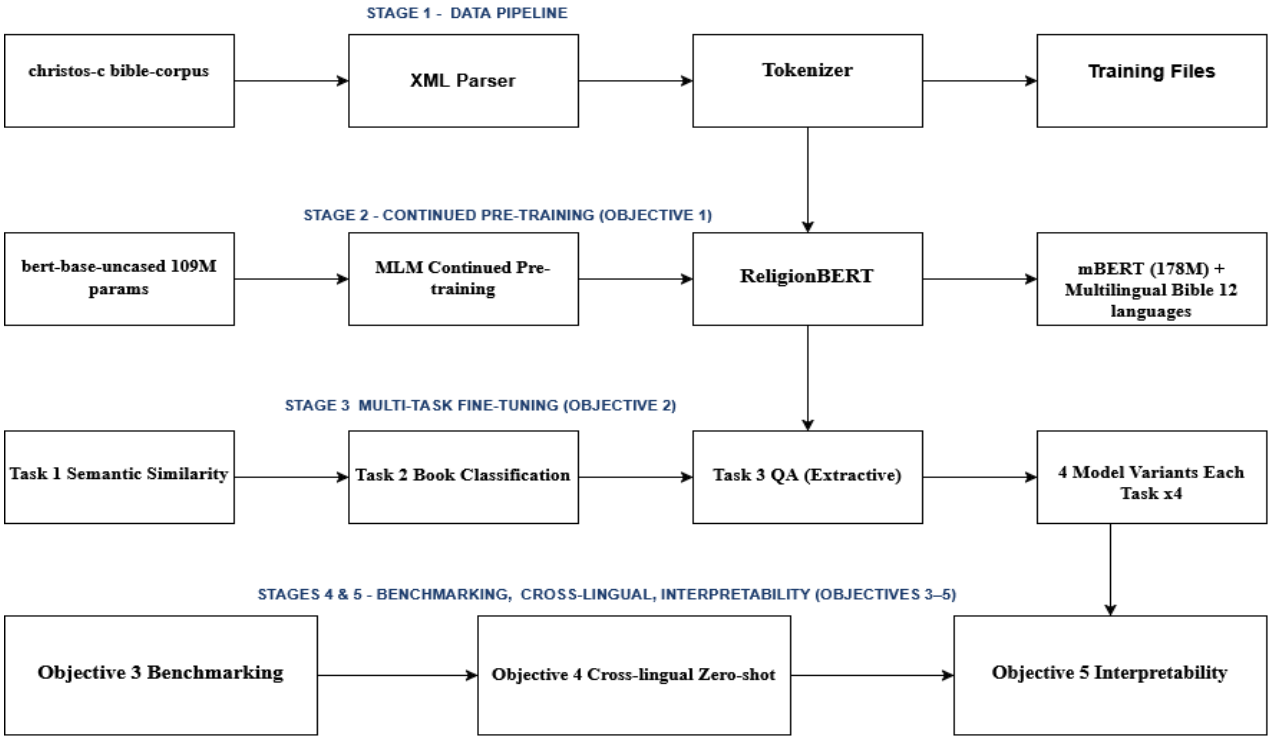

Fig. 3 Full experimental pipeline from corpus collection through domain-adaptive pre-training, multi-task fine-tuning, benchmarking, cross-lingual evaluation, and interpretability analysis.

4.2 Pre-Training Configuration

Both domain-adapted models were trained using the MLM objective with 15% random token masking. Training used the AdamW optimiser with a linear learning rate schedule peaking at 3e-5 with 500 warmup steps and weight decay of 0.01. Batch size was 16 per device with 2 gradient accumulation steps (effective batch 32). Both models were trained for 30,000 steps. All experiments used HuggingFace Transformers 5.0.0 on NVIDIA T4 and A100 GPUs via Google Colab. A custom PermanentDeleteCallback using shutil.rmtree kept only the two most recent checkpoints, preventing storage exhaustion. All trained models were pushed to HuggingFace Hub immediately after training.

Table 5. Hyperparameter configuration for continued pre-training.

Parameter	Value
Training steps	30,000
Effective batch size	32 (16 per device, 2 gradient accumulation steps)
Learning rate	3e-5 (linear warmup 500 steps)
Weight decay	0.01 (AdamW)
MLM masking probability	15%
Precision	FP16 mixed precision
Max sequence length	128 tokens
Checkpoints retained	2 (via PermanentDeleteCallback)

The masked language modelling loss(L_MLM) is expressed mathematically in Equation (1) as

$$L_{MLM} = -E_{x\sim D} \sum_{t\in M} \log P\left(x_t \middle| x_{\setminus M}; \theta\right) \quad (1)$$

where x is an input token sequence drawn from domain distribution D. M is the set of randomly selected masked token positions (|M| / |x| = 0.15). x_t is the original token at masked position t. x\M is the input sequence with masked positions replaced by the [MASK] token, a random token, or the original token according to the 80/10/10 ratio. theta represents the model parameters. P(x_t | x\M; theta) is the model's predicted probability of the original token at position t given the surrounding unmasked context. The loss (LMLM) is calculated as the negative expectation over the biblical training corpus (D), which comprised 388,448 verses across 14 languages. For each training step, the model attempted to maximize the probability (P) of correctly predicting the original token (xt) for all indices in the masked set (M), given the observed context of unmasked tokens (x\M) and the current model parameters (θ).

4.3 Training Dynamics

Figure 4 shows the WandB training dashboard. Figure 5 shows validation loss curves for both models. ReligionBERT's validation loss declined from 1.806 at step 500 to a best of 1.129 at step 28,500 (37.5% total reduction); final checkpoint loss was 1.164. MultiReligionBERT reached a best validation loss of 1.368 at step 29,500, final loss 1.392. The higher absolute loss

for MultiReligionBERT is expected given the greater linguistic diversity of its 12-language training corpus.

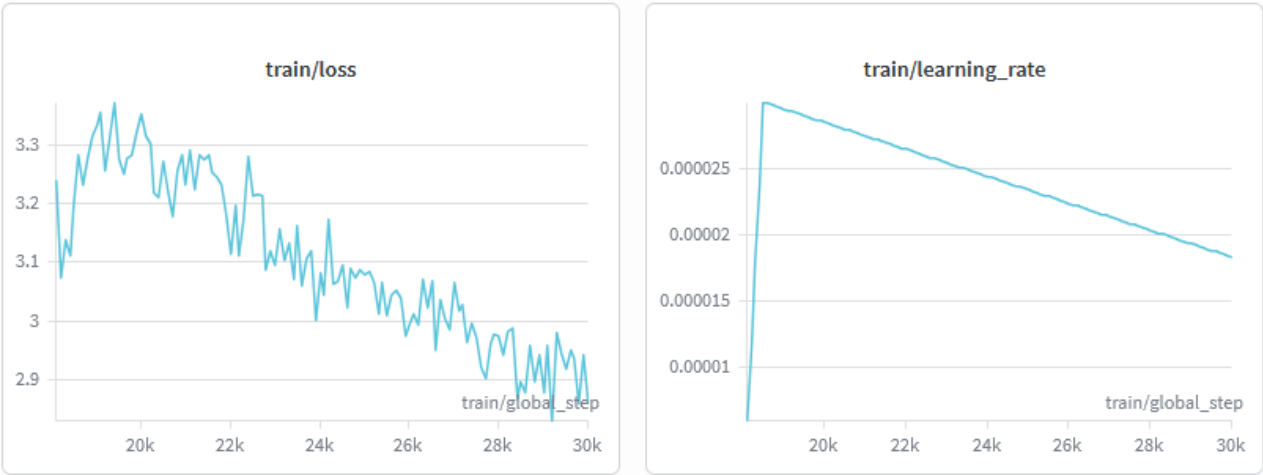

Fig. 4. WandB training dashboard showing ReligionBERT training loss and learning rate schedule across all 30,000 steps.

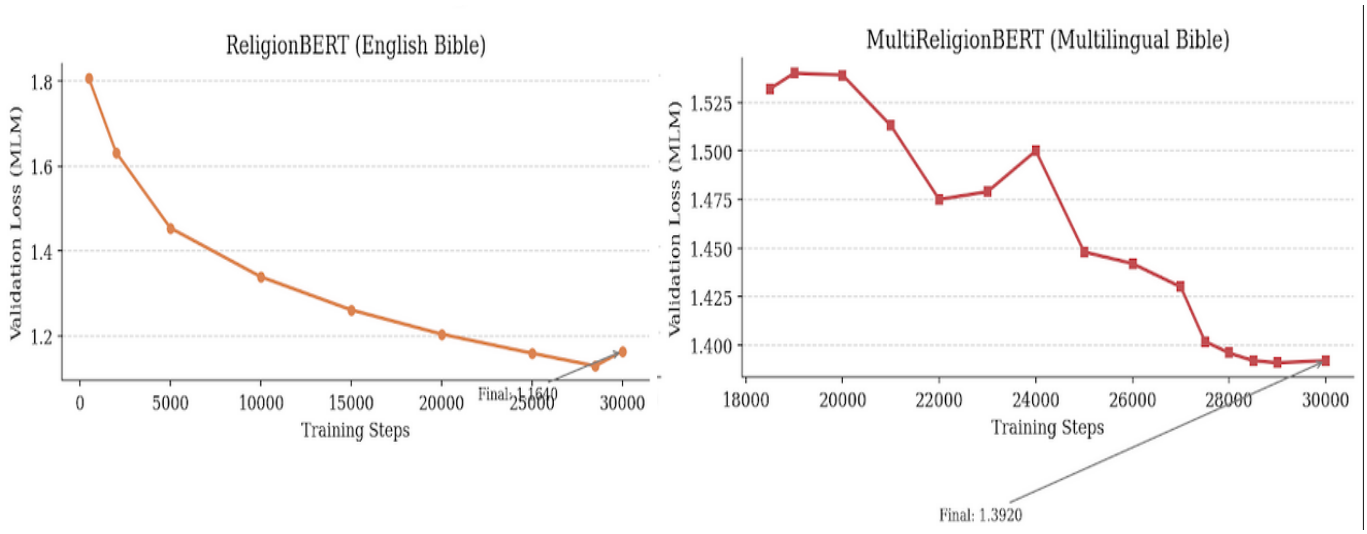

Fig. 5. Validation MLM loss curves. Left: ReligionBERT (final loss 1.164). Right: MultiReligionBERT (final loss 1.392).

Table 6. Selected validation loss values across training.

Model	Step 500	Step 10,000	Step 20,000	Best Loss	Best Step	Final
ReligionBERT	1.806	1.339	1.204	1.129	28,500	1.164
MultiReligionBERT	1.532*	n/a*	n/a*	1.368	29,500	1.392

* Logging began at step 18,500 after session interruptions.

5 Fine-Tuning Methodology

All four model variants were fine-tuned independently on each of the three tasks, producing twelve trained models. Figure 6 illustrates the two-stage training procedure.

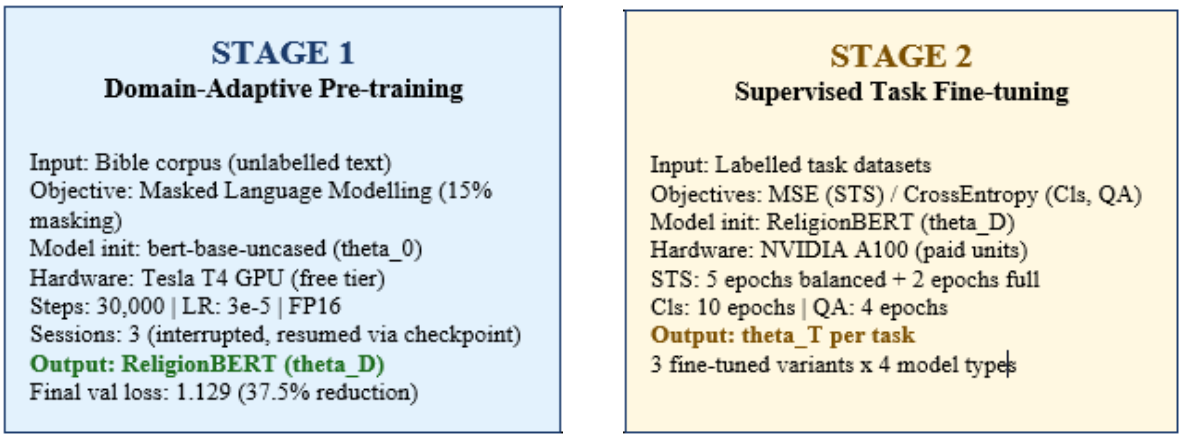

Fig. 6. Two-stage training procedure. Left: Stage 1 domain-adaptive pre-training. Right: Stage 2 supervised task fine-tuning.

5.1 Semantic Similarity. BertForSequenceClassification with num_labels=1 and MSE loss was used. A staged strategy trained for 5 epochs on the balanced 6,392-pair subset at 2e-5, then 2 additional epochs on the full 21,994-pair dataset at 1e-5. Evaluation used Pearson and Spearman correlation.

5.2 Book Classification. BertForSequenceClassification with num_labels=66 and cross-entropy loss was trained for 10 epochs at 2e-5. Macro-averaged F1 was the primary metric to treat all 66 books equally regardless of sample count.

5.3 Extractive QA. BertForQuestionAnswering predicted start and end token positions on inputs tokenised with maximum length 384 and stride 128. Initial 5-epoch runs at 2e-5 produced overfitting after epoch 1 on all models. Hyperparameters were adjusted to 4 epochs, learning rate 1e-5, batch size 8, warmup steps 50. Evaluation used token-level F1 and Exact Match.

6 Experiments and Results

6.1 Perplexity on Held-out Religious Text

Perplexity was computed on 500 held-out English Bible verses by applying 15% random masking and exponentiating the average per-token cross-entropy loss. ReligionBERT achieves 3.73 compared to 15.48 for generic BERT, a 75.9% reduction, confirming successful domain alignment independent of any task-specific fine-tuning.

Perplexity on the target domain is represented mathematically in Equation (11).

$$PPL = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\frac{1}{|M_i|\sum_{t\in M_i}\log P\left(x_t^{(i)}\middle|x_{\setminus M_i}^{(i)};\theta\right)}\right) \quad (2)$$

where PPL is perplexity on the target domain, N is the number of evaluation sequences (500 in this study), M_i is the set of randomly masked positions in sequence i (15% of tokens), and $x_t^{(i)}$ is the original token at masked position t in sequence i. Lower perplexity indicates that the

model assigns higher probability to the observed tokens in religious text, reflecting better domain alignment.

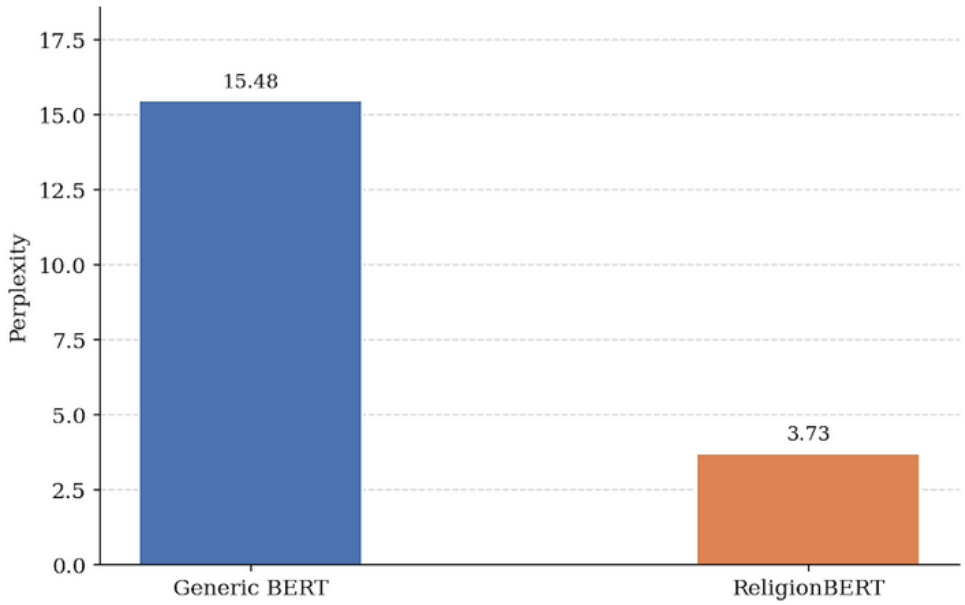

Fig. 7. Perplexity on 500 held-out English Bible verses (lower is better). ReligionBERT (3.73) vs. generic BERT (15.48).

6.2 Semantic Similarity

Table 7 and Figure 8 report test set results. ReligionBERT outperforms generic BERT on both Pearson (+0.0054) and Spearman (+0.0035) correlation. MultiReligionBERT outperforms mBERT on Pearson by 0.0011 but shows a slight Spearman decrease of 0.0103. High base performance above 0.95 Pearson correlation creates a ceiling effect limiting observable gains.

Table 7. Semantic similarity test set results.

Model	Pearson	Spearman	vs. Baseline (Pearson)	vs. Baseline (Spearman)
Generic BERT	0.9569	0.6556	Baseline	baseline
ReligionBERT	0.9623	0.6591	+0.0054	+0.0035
mBERT	0.9624	0.6772	Baseline	Baseline
MultiReligionBERT	0.9635	0.6669	+0.0011	-0.0103

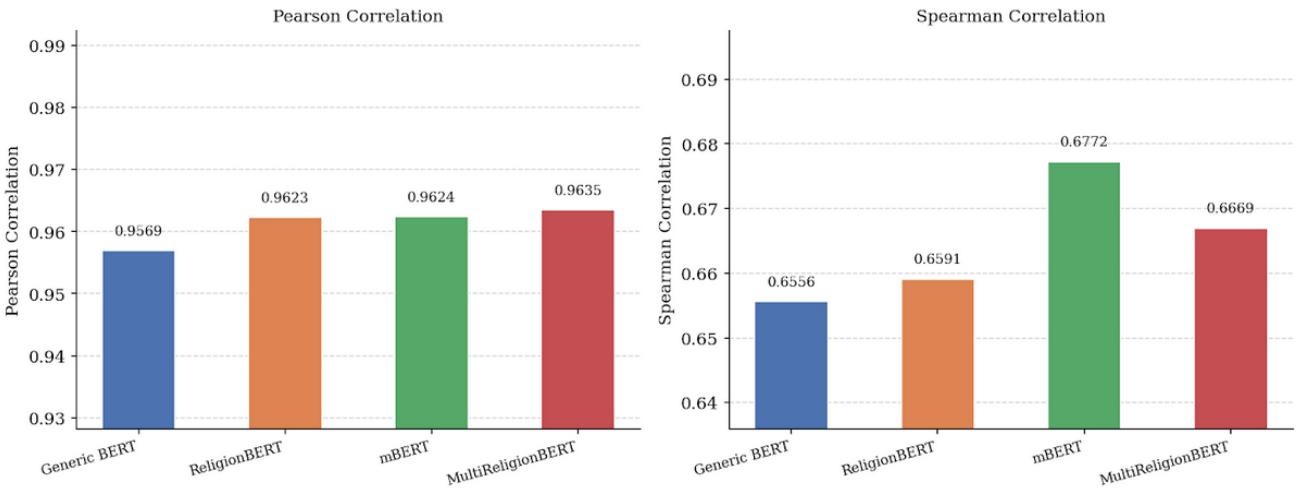

Fig. 8. Pearson (left) and Spearman (right) correlation for all four models on the semantic similarity test set.

6.3 Book Classification

Table 8 and Figure 9 report results on the 66-class task. Random chance yields 1.5% accuracy. ReligionBERT outperforms generic BERT by 2.72 accuracy points and 2.37 macro F1 points. MultiReligionBERT outperforms mBERT by 3.88 accuracy points and 4.21 macro F1 points.

Table 8. Book classification test set results.

Model	Accuracy	Macro F1	Acc vs. Baseline	F1 vs. Baseline
Generic BERT	0.4075	0.3144	baseline	Baseline
ReligionBERT	0.4347	0.3381	+0.0272	+0.0237
mBERT	0.3972	0.2948	baseline	Baseline
MultiReligionBERT	0.4360	0.3369	+0.0388	+0.0421

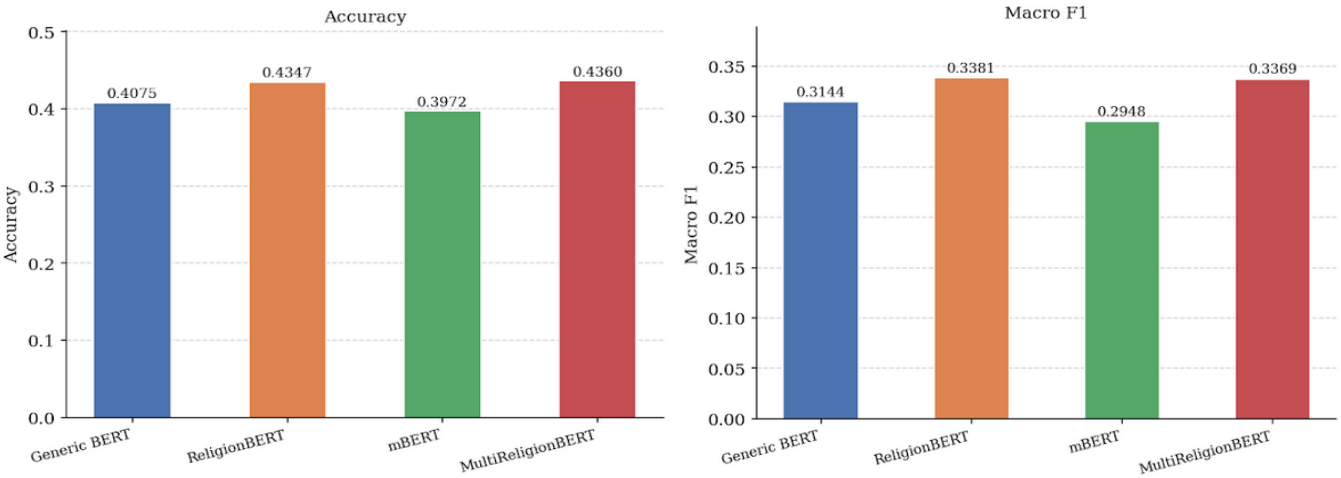

Fig. 9. Accuracy (left) and Macro F1 (right) for all four models on the 66-class book classification test set.

6.4 Extractive Question Answering

Table 9 and Figure 10 report QA test results. ReligionBERT shows the largest gain across all three tasks, improving over generic BERT by 7.50 Exact Match points and 1.57 Token F1 points. This result is consistent with the nature of extractive QA: precise span selection is highly sensitive to token-level contextual alignment, and domain-specific pre-training provides exactly that grounding for religious passages.

MultiReligionBERT shows an unexpected reversal, scoring 7.50 EM points below mBERT. Multilingual domain adaptation shifts English-specific span extraction capabilities in ways that are counterproductive for a small English-only QA dataset, consistent with the known risk of catastrophic forgetting of general-language capabilities during domain-specific continued pre-training.

Table 9. Extractive QA test set results (Exact Match % and Token F1 %).

Model	Exact Match (%)	Token F1 (%)	EM vs. Baseline	F1 vs. Baseline
Generic BERT	32.50	58.91	baseline	Baseline
ReligionBERT	40.00	60.48	+7.50	+1.57
mBERT	48.33	70.54	baseline	Baseline
MultiReligionBERT	40.83	65.39	-7.50	-5.15

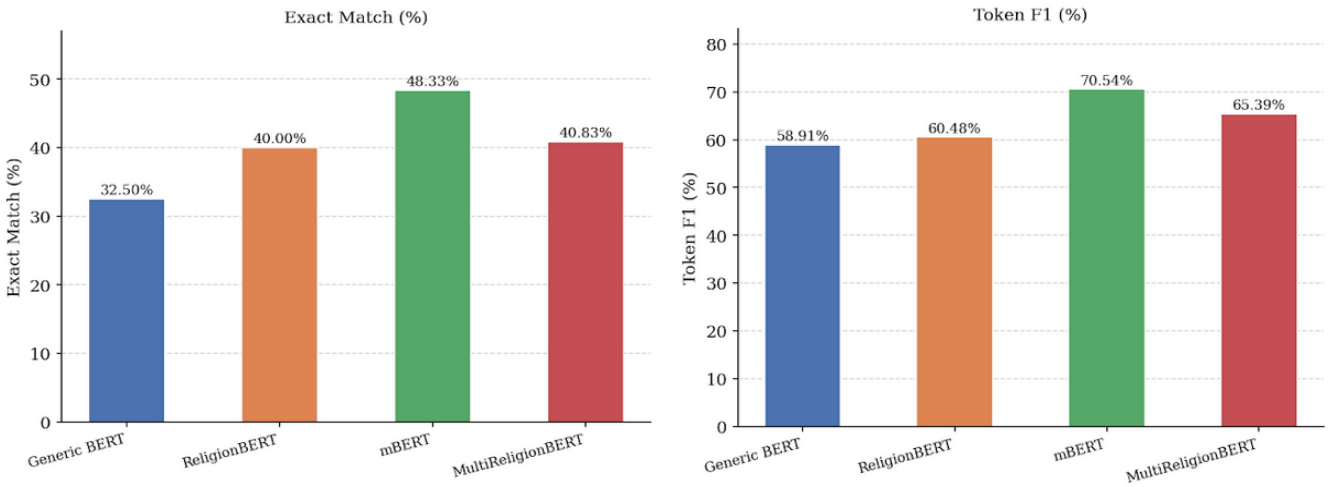

Fig. 10. Exact Match (left) and Token F1 (right) for all four models on the extractive QA test set.

6.5 Benchmarking Synthesis

Table 10 consolidates all results. ReligionBERT outperforms generic BERT on five of six metrics. MultiReligionBERT outperforms mBERT on four of six. The domain adaptation advantage is strongest on QA for the English model (+7.50 EM) and on classification for the multilingual model (+3.88 accuracy points). Figures 11, 12, and 13 show the normalised comparison, radar chart, and improvement heatmap.

Table 10. Complete benchmarking results across all tasks and models.

Model	Sim Pearson	Sim Spearman	Cls Acc	Cls F1	QA EM%	QA F1%
Generic BERT	0.9569	0.6556	0.4075	0.3144	32.50	58.91
ReligionBERT	0.9623	0.6591	0.4347	0.3381	40.00	60.48
mBERT	0.9624	0.6772	0.3972	0.2948	48.33	70.54
MultiReligionBERT	0.9635	0.6669	0.4360	0.3369	40.83	65.39

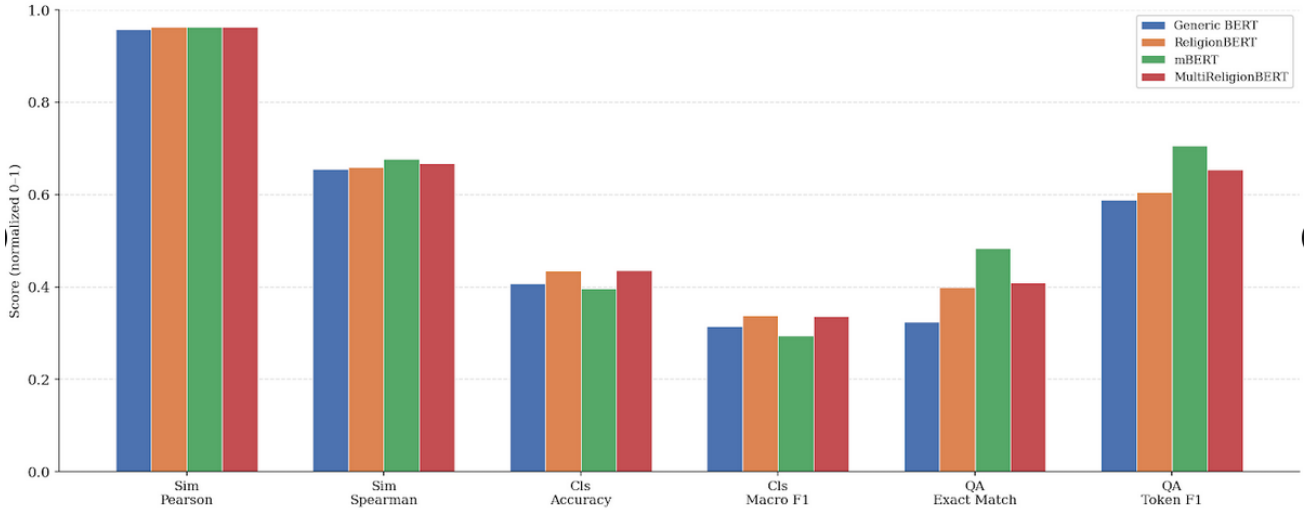

Fig. 11. All four models across all six evaluation metrics on a normalised 0-1 scale.

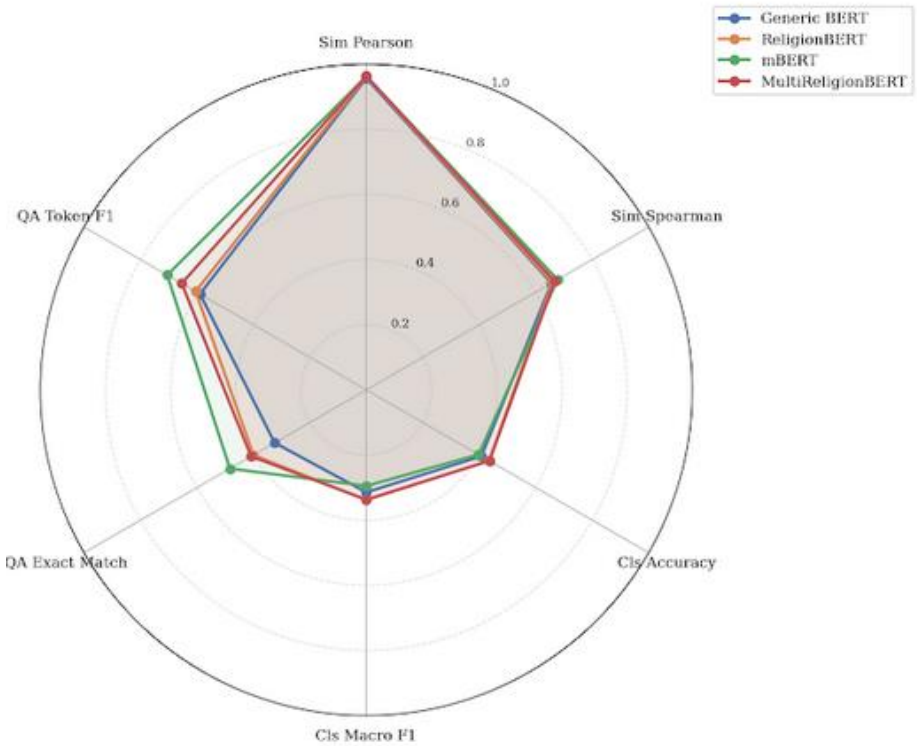

Fig. 12 Radar chart across all six metrics.

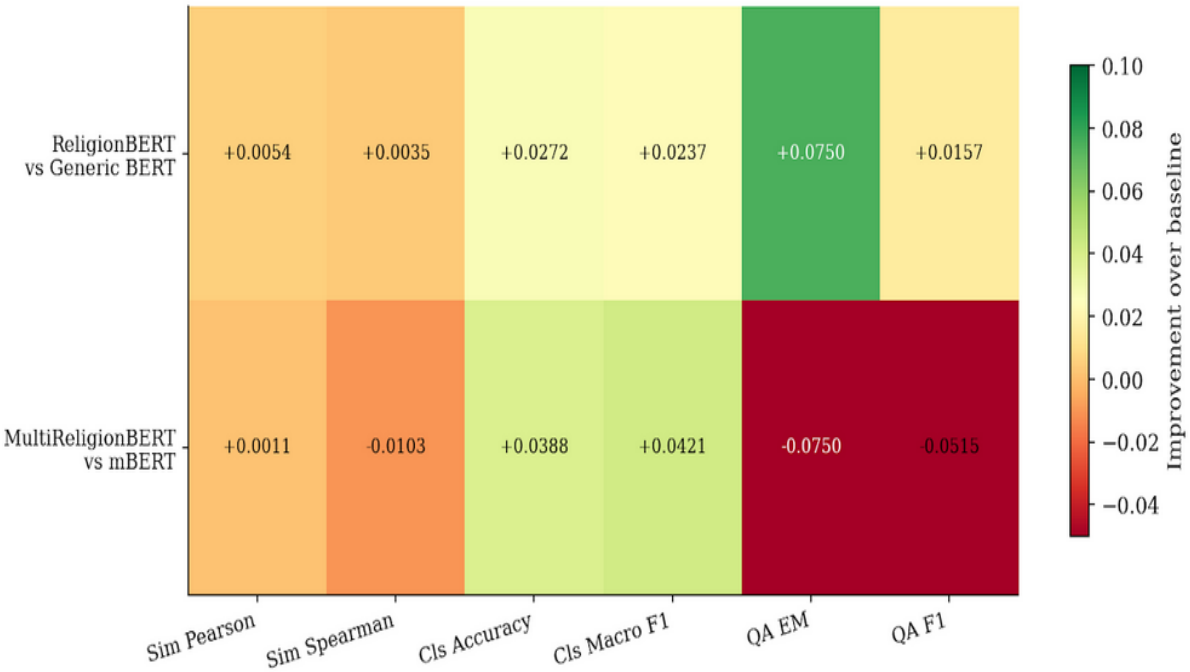

Fig. 13 Improvement heatmap of domain-adapted vs. generic models. Green = gain; red = loss

7 Cross-Lingual Zero-Shot Transfer

Three multilingual models (mBERT, MultiReligionBERT, and XLM-RoBERTa [7]) were evaluated on zero-shot book classification applied to five African language Bible corpora. All models were fine-tuned exclusively on English classification data; no target-language fine-tuning was performed. XLM-RoBERTa was fine-tuned on the English training set first, reaching 35.83% test accuracy and 0.2617 macro F1. Evaluation sets were constructed by sampling approximately four verses per available book in each target language. Table 11 reports full results.

Table 11. Zero-shot cross-lingual book classification results on five African languages.

Language	Family	Samples	mBERT Acc	MultiRel Acc	XLM-R Acc	mBERT F1	MultiRel F1	XLM-R F1
Amharic	Semitic	264	0.0152	0.0152	0.1326	0.0005	0.0005	0.0886
Shona	Bantu	252	0.0556	0.0714	0.0556	0.0267	0.0402	0.0381
Xhosa	Bantu	264	0.0189	0.0227	0.0530	0.0040	0.0072	0.0396
Ewe	Gbe	297	0.0000	0.0337	0.0034	0.0000	0.0187	0.0011
Swahili	Bantu	286	0.0280	0.0559	0.0874	0.0098	0.0262	0.0364

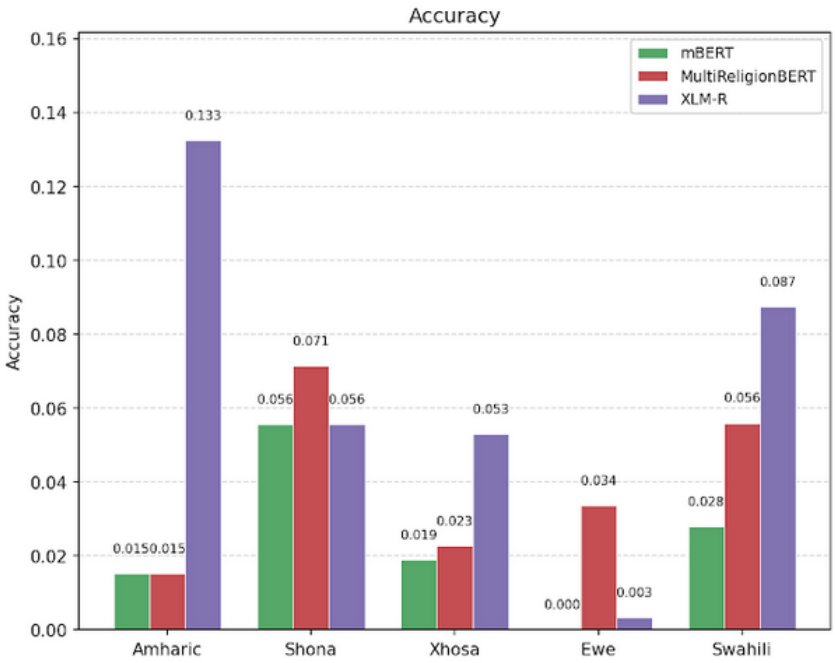

Fig. 14. Zero-shot accuracy for mBERT (green), MultiReligionBERT (red), and XLM-RoBERTa (blue) across five African languages.

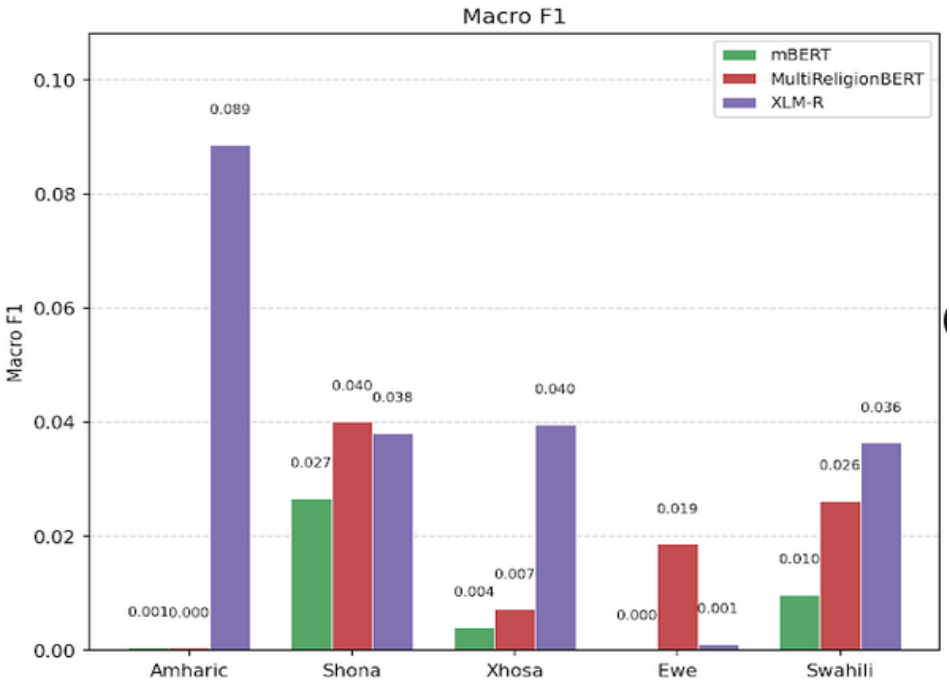

Fig. 15 Macro F1 across five African languages

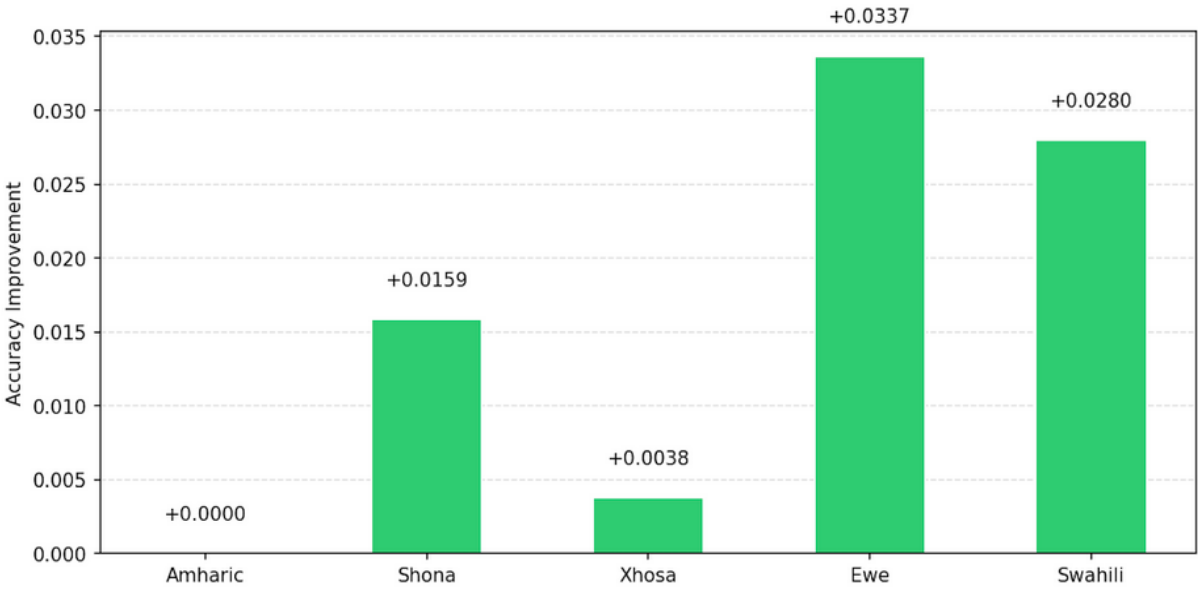

Fig. 16 MultiReligionBERT accuracy gain over mBERT per language.

MultiReligionBERT outperforms mBERT on four of five languages. The most striking result is Ewe: mBERT predicts zero correct book labels across all 297 test samples while MultiReligionBERT achieves 3.37% accuracy. Ewe is severely low-resource in standard multilingual corpora but is covered by a New Testament translation in the Bible pre-training data. XLM-RoBERTa leads on Amharic, Xhosa, and Swahili due to its larger general pre-training corpus, but trails MultiReligionBERT on Ewe, where domain-specific religious corpus coverage provides an advantage that general-purpose scale alone does not match.

8 Discussion

8.1 Why Extractive QA Benefits Most

The largest single gain is ReligionBERT's 7.50 Exact Match improvement on extractive QA. Span-selection tasks require the model to maintain precise token-level alignment between a question and passage context. General-purpose BERT allocates representational capacity to contemporary secular language; ReligionBERT redistributes that capacity toward the vocabulary, syntax, and semantic patterns of Biblical text, which is most consequential for tasks demanding fine-grained span matching.

8.2 Ceiling Effects on Semantic Similarity

The modest similarity gains reflect a ceiling effect inherent in the dataset design. The dominant class consists of cross-translation pairs of identical verses, and any BERT variant fine-tuned on sufficient such pairs converges to high Pearson correlations. The Spearman metric is more sensitive because it captures rank-order agreement and is more responsive to errors in the mid and low similarity ranges.

8.3 The LLM-Assisted Pipeline as a Contribution

The QA dataset construction pipeline represents a distinct methodological contribution. For specialised domains with no existing annotated benchmark, LLM-assisted annotation with structured prompting, automated substring validation, and human spot-checking provides a

practical path to task-specific datasets at a fraction of the cost of fully manual annotation. The 1,199-example dataset achieved 100% verified quality. The approach is directly transferable to other sacred text traditions such as the Quran or Buddhist sutras.

8.4 Cross-Lingual Transfer Analysis

The cross-lingual results demonstrate that domain-specific multilingual pre-training on religious text provides transferable advantages, even in zero-shot conditions. The Ewe result is the most compelling: mBERT's complete failure (0.00%) and MultiReligionBERT's non-trivial performance (3.37%) reveal that inclusion of a language in the religious pre-training corpus provides transfer signal that mBERT's general multilingual pre-training cannot supply. This has implications for low-resource African language NLP more broadly, as Bible translations provide one of the most consistent multilingual text resources across African languages.

8.5 Limitations

Several limitations bound this study. The pre-training corpus is limited to the Bible; religious NLP encompasses broader traditions including the Quran, Vedas, and Buddhist sutras, each with distinct linguistic properties. The QA dataset is small at 959 training examples, near the lower bound for reliable BERT fine-tuning, as evidenced by the overfitting in initial runs. The cross-lingual evaluation is restricted to book classification, which may not fully represent practically relevant cross-lingual capabilities.

9 Conclusion

We have presented ReligionBERT and MultiReligionBERT, two domain-adapted BERT models produced by continued MLM pre-training on Biblical corpora. Benchmarking across three tasks and four model variants shows consistent improvements over generic baselines: ReligionBERT outperforms generic BERT on five of six metrics, with a peak gain of +7.50 Exact Match points on extractive QA, and perplexity on held-out religious text falls by 75.9%. Zero-shot cross-lingual evaluation across five African languages demonstrates that multilingual domain-specific pre-training confers transferable advantages, most strikingly for the severely low-resource Ewe where mBERT achieves 0.00% and MultiReligionBERT achieves 3.37%. Both models and all datasets are publicly available on HuggingFace. Interpretability analysis revealing the internal representational mechanisms behind these performance gains is reported in a companion paper.

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